Ultrainfinitism II:
Infinite Wholes and the Boundary of Effing

Sinkless collapse, labelled plurality, pointless algebras,

bridges, two ineffabilities, and transforms of the transfinite

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Abstract

The first essay read the finite as a principal-ultrafilter shadow of an infinite One and anchored the reading in Łoś's theorem, credal envelopes and Stone duality. This sequel sharpens the principle and tests it against new machine-checked results. *Infinite wholes are primary; finite information appears only as observation of them*—a prefix read below a depth, an outcome under a budget, a coordinate, a clopen—*never as their parts.* Four results carry the sharpened principle. With no empties anywhere, extensional structure yields exactly one thing, Aczel's $\Omega = \{\Omega\}$. Plurality needs labels: two labels already give uncountably many leafless wholes, only countably many of which have a finite presentation, and no bounded observation tells them all apart. A single whole can carry uncountably many self-identifications. A complete atomless algebra has no points at all, so individuals appear only when observation is cut down to finite joins and a free choice is made. Models assembled from finite pieces—the cofinite core of the free ultrafilters, $\mathcal{P}(\mathbb{N})/\mathrm{fin}$, hyperfinite runs—are treated explicitly as *bridges*, not foundations. We then separate three senses of "without normal forms", order what can be *effed* (finitely and checkably presented) in a hierarchy tied to the outcomes of bounded search, and distinguish two ineffabilities: one from symmetry (no deterministic equivariant choice among indiscernibles) and one from self-reference (Lawvere's fixed-point theorem). Transforms of infinite trees and runs converge, conserve their charges, fold silently and pump without bound, each under stated hypotheses. The diagonal and symmetry theorems are finite, checked statements; they eff exactly where effing must stop. Every result stated as proved names its Lean declaration; readings and bridges are labelled as such.

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1 Introduction

The first essay [1] argued that only the infinite is truly real and that every finite object is a *simulation*: the One read at a single coordinate by a principal

ultrafilter. Its Law of Oneish said that beneath the carvings there is one thing, and that carvings are how a perspective makes a finite world to live in. Two bridge principles graded the step from phenomenology to ontology: *No-Atom* (if no boundary is absolute, there is no minimal, fully separated individual) and *Priority of the Whole* (a carving presupposes the manifold it carves, not conversely). The mathematical home was the ultraproduct $(\prod_i M_i)/\mathcal{U}$ with Łoś's theorem as the law of transfer, indexed explicitly as a relative One $\mathrm{One}_{(M,\mathcal{U})}$; the native home was atomless (gunky) mereology, welded by Stone duality to perfect point-spaces in which no finitary point stands alone. Its "Honest scope" remark kept proved results apart from the envisioned, and its Lean appendix transcribed the anchors.

The sharpened principle. The first essay still let the finite appear as a coordinate *of* a product—one factor among the M_i. Read strictly, that concedes too much: a product is assembled from its factors. The sequel takes the opposite order of construction as primary.

Principle 1 (wholes first)**.** Infinite wholes are primary. Finite information appears only as *observation* of a whole: a prefix (agreement below a depth), a budget (the outcome of a search cut off at depth n), a coordinate (a principal ultrafilter), a clopen (a finite join of basic observations). It never appears as a *part* from which the whole is assembled.

The difference is the difference between an initial algebra, whose elements are built from finite parts, and a final coalgebra, whose elements are known by their finite observations [34]. A model assembled from finite pieces can still be useful, and several are used below. The algebra $\mathcal{P}(\mathbb{N})/\mathrm{fin}$, for instance, is atomless and has a perfect Stone space, and so is a faithful image of the thesis; but its zero *is* the class of all finite sets. Such a model is a *bridge*: it transports truth between finite observations and wholes, but it is not the foundation, because it is made out of what the foundation is supposed to make.

Effing. We say that a thing is *effed* when a finite, checkable presentation determines it: a finished certificate, a finite graph, a program, an approximation scheme with a named limit. The ineffable is what no such presentation reaches. The paper's second half orders the grades of effability and shows that the places where effing fails are themselves sharply effable.

What is new.

1. *The One by structure* (Section 2): sinkless collapse to Ω (Theorem 2); plurality only through labels, uncountable and mostly without finite presentation (Theorem 4); one whole with uncountably many self-identifications (Theorem 6); a complete atomless algebra with no points (Theorem 8); opaque atoms, the "empty" as a whole seen from outside (Theorem 10); and finite talk as distinction, which induces finite quotients of a whole rather than finite parts (Section 2.6).

2. *Bridges* (Section 3): what every free One affirms is exactly what holds cofinitely (Bridge 14); no ultrafilter is fixed by every symmetry, in one form shared with an observer theorem (Bridge 16); and $\mathcal{P}(\mathbb{N})/\mathrm{fin}$ (Bridge 17).

3. *Three senses of "without normal forms"* (Section 4), including Lawvere's theorem for types and the collapse of every arrow model of rational types (Theorems 19–20).

4. *A hierarchy of effability* tied to the outcomes of bounded search (Section 5, Theorem 21).

5. *Two ineffabilities* (Section 6): symmetric and diagonal.

6. *Transforms over transfinite entities* (Section 7): ω-limits (Theorem 25), hyperfinite runs (Bridge 26), the quiet fold (Theorem 27) and the affine pump (Theorem 29).

Conventions. A result is stated as a *theorem* only when a named Lean 4 declaration proves it without `sorry`; the theorem heading names the declaration and its module. Results numbered as a *Bridge* are proved in the same sense. Module paths are repository-relative; we abbreviate `Mettapedia/Logic/Metaphysics/X.lean` as `Metaphysics/X.lean` and `Mettapedia/Languages/ProcessCalculi/ConstraintMultisetRewriting/Step/X.lean` as `Step/X.lean`. Unless stated otherwise the anchors depend on at most the standard axioms `propext`, `Classical.choice` and `Quot.sound`; where a result uses less, we say so. A *bridge* is a model built from finite pieces and used to transfer truth. A *reading* is an interpretation of a theorem, offered as interpretation; nothing in a reading is claimed as proved.

2 The One by structure; plurality by carving

2.1 Sinkless collapse: no empties, one thing

Work with hypersets: sets under Aczel's anti-foundation axiom, in which every graph has exactly one decoration and equality is bisimilarity [3, 5]. In the development the axiom is a theorem about a model of hypersets, and $\Omega = \{\Omega\}$ is the hyperset pictured by a one-node loop. Call a hyperset x *hereditarily nonempty* when every member of its transitive closure, x itself included, has a member.

Theorem 2 (sinkless collapse; `hereditarilyNonempty_iff_eq_quineAtom`, `eq_of_hereditarilyNonempty`, `decorate_eq_quineAtom_iff_sinkless`, `Metaphysics/UltrainfiniteSinkless.lean`)**.**

1. *A hyperset is hereditarily nonempty if and only if it is* Ω*. In particular any two hereditarily nonempty hypersets are equal.*

2. *A graph pictures* Ω *at a node exactly when no sink (node without children) is reachable from that node.*

3. Control *(`sink_plurality`): with a sink, plurality appears:* $\varnothing \neq \{\varnothing\}$*, and neither is hereditarily nonempty.*

The proof exhibits "both are hereditarily nonempty" as a bisimulation of the membership graph (`isBisimulation_hereditarilyNonempty`) and applies strong extensionality. Parts 1–2 use no choice.

The collapse is a consequence of *extensional bisimulation*: hypersets with bisimilar membership graphs are equal. It owes nothing to any principle about identity proofs. Nor is the surviving whole empty: Ω has a member, namely itself, and every member of its closure is Ω (`eq_quineAtom_of_inClosure`).

Every sinkless presentation pictures the same whole, whatever its shape: the loop, the two-cycle, and the full binary tree below are three pictures of one thing (Figure 1).

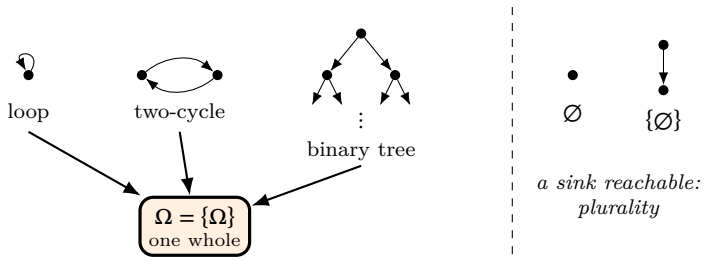


Figure 1: Sinkless collapse (Theorem 2). Every sinkless graph pictures Ω at every node; plurality of pure sets begins only where a sink—an empty—is reachable.

Reading 3 (the Law of Oneish, by structure). In well-founded set theory every set reaches $\varnothing$, so the empty set is where all distinctions among pure sets bottom out. Theorem 2 supplies the complement: once no empty is reachable anywhere, extensional structure has exactly one thing to say. Pure plurality is *sourced* in emptiness. An ultrainfinitist who denies finite atoms and empty individuals therefore cannot get plurality from pure membership at all; it must come from somewhere else. The next subsection shows where.

2.2 Distinctions need labels: leafless wholes

Read a tree by its labels at paths, as a function from finite lists of child indices to labels. A tree is *leafless* when it is well formed and every label present in it has a child; so it has no variable or constant leaves and is built from function symbols of positive arity alone.

Theorem 4 (plurality by labels; `Metaphysics/UltrainfiniteLeafless.lean`)**.** *Let* σ *be a signature.*

1. No leafless tree is finite *(`Leafless.infinite_support`, `Leafless.infinite_support_sub`, `not_leafless_ofTerm`): it is present at infinitely many paths, so is every subtree, and no finite term reads as one.*

2. Two symbols give uncountably many *(`not_countable_setOf_leafless`): if $f \neq g$ have positive arity, the leafless trees are uncountable.*

3. Few have a finite presentation *(`countable_leafless_rational`, `exists_leafless_not_rational`): over countably many symbols the rational leafless trees (finitely many distinct subtrees) are countable, each the unfolding of a finite code; hence some leafless tree is not rational.*

4. No bounded observation decides identity *(`exists_ne_agree_below`): for every depth d there are two distinct leafless trees with the same label at every path shorter than d.*

5. Control: one symbol, one whole *(`existsUnique_leafless_of_subsingleton`): with at most one function symbol, and that of positive arity, there is exactly one leafless tree.*

The hypothesis of part 2, two distinct symbols of positive arity, is minimal, as part 5 shows. With one unary symbol the only leafless tree is $f(f(f(\cdots)))$, whose membership reading is again Ω. So Theorems 2 and 4 together locate plurality exactly: without sinks and without a second label, there is one whole; a second label, and no leaf at all, already gives a continuum of wholes.

Reading 5 (plurality by carving). Labels are the first carving. The wholes they distinguish are not assembled from labelled finite pieces: parts 1 and 4 say there are no finite pieces to assemble from, and no finite inspection closes the question of which whole is in view. Part 3 says that most of these wholes cannot be named by a finite graph at all. Plurality is real, but it is plurality of infinite wholes, carved by labels and observed by prefixes.

2.3 One whole with inner diversity

The full binary tree has the finite Boolean lists as nodes and an edge from p to $p +\!\!+ [b]$. Each subtree below the root is again the full binary tree (`child_cons_iff`), so as an iterated multiset it is $T = \{T, T\}$. It is sinkless.

Theorem 6 (one with inner diversity; `inner_diversity`, `Metaphysics/UltrainfiniteInnerDiversity.lean`)**.**

1. *The edge-preserving permutations of the nodes form a subgroup `treeAut`, and it is uncountable (`not_countable_treeAut`): choosing at each node whether to swap its children is an automorphism (`swapPerm_mem_treeAut`), and distinct choices give distinct automorphisms (`swapPerm_injective`).*

2. Control: as a set it is one thing. *The tree pictures* Ω *at every node (*`decorate_binaryTree`*), and every automorphism of the membership structure of hypersets is the identity (*`eq_self_of_mem_automorphism`*).*

Reading 7 (connected, not contractible). As an object, T is one: read extensionally it is Ω, and Ω has exactly one self-identification. As a *structure*, T has uncountably many. In the vocabulary of homotopy type theory [24] this is the profile of a type that is connected (there is one object up to identification) but not contractible (its self-identifications do not all coincide). The groupoid model of Hofmann and Streicher is the classical place where identity carries such content [23]. We offer this only as a reading of one object with an uncountable automorphism structure, and the reading rests on Theorem 6, not on any choice of identity principle. In particular it is not a consequence of dropping Streicher's axiom K: failure of uniqueness of identity proofs does not by itself give connectedness, nontrivial loops, or the equality of all objects, which are separate properties. Lean's equality is proof-irrelevant and validates K, so it cannot itself carry a nontrivial group of self-identifications; the development therefore carries identity as explicit structure, the subgroup `treeAut`, rather than as equality. The theorem is about that structure; the homotopical vocabulary is interpretation.

2.4 Stone points against locale points

The first essay proved that a Boolean algebra is atomless exactly when its Stone space is perfect (`isGunky_iff_perfect_stoneSpace`, `Mettapedia/Logic/StoneGunkDuality.lean`). A perfect Stone space still has points: ultrafilters exist in every nontrivial Boolean algebra by the Boolean prime ideal theorem [37, 25], a form of choice. Ultrafilters respect only *finite* joins. Ask instead for the points of the algebra as a locale—completely prime filters, which respect *all* joins—and the answer changes.

Theorem 8 (a complete atomless algebra has no points; `exists_atom_of_point`, `isEmpty_point_of_isGunky`, `isEmpty_point_regular_opens`, `Mettapedia/SetTheory/CarveOuts/HeytingValued/Gunky.lean`)**.**

1. *In a complete Boolean algebra every point (completely prime filter) is generated by an atom. Hence a complete atomless Boolean algebra has no point at all.*

2. *For a Hausdorff space with no isolated point, the regular open sets form an atomless complete Boolean algebra with no point. The frame of all open sets of a* T_1 *space with no isolated point is atomless, has a point for every point of the space, and none of its points is principal (*`not_isPrincipal_opens`*).*

3. *Cantor space and the real line are witnesses (*`cantorGunkyWitness`*,* `realGunkyWitness`*).*

The three cases line up as follows.

Algebra	complete?	atoms	Stone points (ultrafilters)	locale points
$\mathcal{P}(\mathbb{N})$	yes	singletons	the coordinates, isolated; free ones by choice	the coordinates only
$\mathcal{P}(\mathbb{N})/\mathrm{fin}$	no	none	by choice only; none isolated	— (not a frame)
regular opens of $2^{\mathbb{N}}$	yes	none	by choice only; none isolated	none at all

Table 1: Points, three ways. Atoms of $\mathcal{P}(\mathbb{N})$ are the singletons (the first essay's `isAtom_set_singleton`, `not_isGunky_set`); its locale points sit at atoms by Theorem 8, part 1. The second row is Bridge 17. In the third row the locale points are Theorem 8, part 2, and "none isolated" is the first essay's duality applied to an atomless algebra.

Reading 9 (the pointless One). The complete atomless One has no individuals at all. Individuals appear only when observation is cut down to finite joins (passing from a locale to a Boolean algebra of clopens) *and* a free choice is made (an ultrafilter). This is the home Johnstone argued for in "the point of pointless topology" [26], here doing ultrainfinitist work: a point is an act of observation with a choice in it, not a constituent of the space.

2.5 Opaque atoms: an empty is a whole seen from outside

The first essay's objection section asked what becomes of the empty set and the urelement in an infinite-native foundation. A many-coloured construction of set universes gives one answer. Each colour builds sets of its own objects, and every colour it sees contributes its objects as *opaque atoms*: references that keep their identity but expose no membership.

Theorem 10 (reference without containment; `atom_memberless`, `atom_injective`, `atomEmbedding`, `transparent_reflection_impossible`, `Mettapedia/SetTheory/ColoredOpaqueUniverses.lean`)**.** *An atom of colour* c *whose payload is an object of colour* d *has no members of colour* c*, whatever the payload holds; yet distinct payloads give distinct atoms, so the whole universe of* d *embeds into that of* c *as atoms. A* transparent *reflection—an injective encoding of a universe's own powerclass into itself—is impossible (Cantor).*

Reading 11 (the empty as an outside view). From inside colour c, an atom looks exactly like an empty: it has no members. From colour d it is a whole object with all its structure. "Empty" is thus a property of the *view*, not of the thing. Opacity is what separates mutual reflection from unrestricted comprehension: the veil lets every universe contain every other as references, while Cantor's

diagonal forbids any universe to contain itself transparently. The construction is stated at a finitary, well-founded syntactic tier; its coalgebraic, non-well-founded version is not claimed here.

2.6 Finite talk as distinction, in the shadow of the infinite

What, then, does finite talk about a whole do, if it never names a finite part? It draws distinctions. A yes/no observation is an element b of a Boolean algebra B of observations: "inside b" or "inside its complement". Finitely many observations $b_1, \ldots, b_n$ generate a finite subalgebra $B_0 \subseteq B$. Every finite Boolean algebra is atomic and isomorphic to the powerset of its atoms; the atoms of B_0 are the nonzero meets $\pm b_1 \wedge \cdots \wedge \pm b_n$, there are at most 2^n of them, and they partition the top [18]. Dually, B_0 is a partition of the Stone space $S(B)$ into finitely many clopens, that is, a continuous surjection of $S(B)$ onto a finite discrete space; and $S(B)$ is the inverse limit of these finite quotients over the finite subalgebras of B [25].

When B is gunky, these atoms belong to the shadow, not to the whole. B has no atoms at all, and its Stone space has no isolated point (`isGunky_iff_perfect_stoneSpace`); so each atom of B_0 splits again in B, and each cell of the finite partition contains infinitely many points. Finitely many distinctions never pick out a finite *part* of a whole; they induce a finite *quotient* of it, a shadow.

Theorem 12 (finite shadows; `Metaphysics/UltrainfiniteShadows.lean`, `Metaphysics/UltrainfiniteShadowsStone.lean`, `Metaphysics/UltrainfiniteShadowsCantor.lean`)**.**

1. *Finitely many distinctions generate a finite subalgebra (`finite_closure`) whose atoms, the* shadow atoms*, partition the top (`isLUB_shadowAtoms`); the whole is the directed union of its finite shadows (`iUnion_shadows_eq_univ`).*

2. *An algebra is gunky exactly when no shadow atom of any finite shadow is an atom of the whole (`isGunky_iff_forall_shadowAtoms_not_isAtom`); in a gunky algebra every shadow atom splits in a finer shadow (`exists_finer_shadow_split`); a finest shadow exists exactly when the algebra is finite (`finite_iff_exists_shadow_forall_isAtom`).*

3. *Dually, a finite shadow is a finite clopen quotient of the Stone space; a shadow atom is an atom of the whole exactly when its class is a single point (`isAtom_iff_fibre_eq_singleton`), and in a gunky algebra every class is infinite (`shadowMap_fibre_infinite`).*

4. *For Cantor space, the space is the inverse limit of its depth-n shadows (`shadowLimitHomeomorph`, with the universal property `existsUnique_lift`); each class is homeomorphic to the whole (`fibreHomeomorph`); and finitely many clopen distinctions never single out a point (`exists_ne_of_finite_distinctions`).*

The inverse-limit presentation of a general Stone space by its finite shadows is not yet formalized (Section 9).

Tolerances as observers. The distinction calculus, already formalized in the development, reads an observer the same way, with grades [19]. A *tolerance* is a bounded, reflexive, symmetric similarity on states, not assumed transitive; its complementary distance may be zero on distinct states. For a tolerance satisfying the triangle law, zero distance is an equivalence, and the observer it induces is the quotient map: two states look the same to it exactly when the tolerance cannot tell them apart (`quotientObserver_equal_iff`, `Mettapedia/Cybernetics/DistinctionCalculus/Observation.lean`). A finite report induces the crisp case: indistinguishable exactly when the reports agree (`ofReport_indistinguishable`, `Mettapedia/Cybernetics/DistinctionCalculus/Basic.lean`). In both cases the observer's world is a quotient of the states, never a part of them.

Computation in the same picture. A computable function on infinite objects—streams, Cantor space, infinite trees—is continuous: each output distinction depends on finitely many input distinctions [41]. On Cantor space the finite-interaction form is formalized: a map into a discrete space is continuous exactly when it depends only on a fixed finite prefix, and exactly when a finite decision tree computes it (`continuous_iff_exists_depth`, `continuous_iff_exists_decisionTree`, `Metaphysics/UltrainfiniteShadowsCantor.lean`); a uniform oracle notion of computability is not formalized. An observer of the distinction calculus that resolves only to depth n has at most 2^n classes, all infinite (`ResolvesWithin.finite_quotient`, `ResolvesWithin.infinite_class`, `Metaphysics/UltrainfiniteShadowsDistinction.lean`). The productive transforms of Section 7.1 are an instance by definition: an output label below depth $d+1$ depends only on input labels below depth d (`Productive`, `Step/TransfiniteLimits.lean`).

Reading 13 (the computable as a shadow). The computable is the shadow that finite interactions with infinite wholes leave behind. A program consults finitely many distinctions of its input before committing to each distinction of its output, so what it computes factors, at every stage, through a finite quotient of the input. The "computable" rung of the effability hierarchy (Section 5) is in this sense the rung of continuous shadows.

3 Bridges: the indexed One

The results of this section are *bridges*. Each concerns models defined by finiteness over coordinates—cofinite sets, sets modulo finite differences, ultrapowers of families—and the free ultrafilters they mention exist only by choice. They transport truth between observation and whole; they do not ground the whole.

3.1 B1: the cofinite core and Łoś

Call an ultrafilter *free* (`IsFree`) when it is the principal ultrafilter of no point. For ultrafilters this is the same as refining the cofinite filter (`isFree_iff_le_cofinite`) and the same as containing no finite set (`isFree_iff_finite_notMem`); free ultrafilters exist exactly on infinite index sets (`exists_free_iff_infinite`). Write UltraTrue($\mathcal{U}, P$) for "P holds at $\mathcal{U}$-almost every coordinate", the verdict of the relative One of the first essay.

Bridge 14 (the effable core of the free One; `Metaphysics/UltrainfiniteCofinite.lean`, `Metaphysics/UltrainfiniteCofiniteLos.lean`)**.** *For every index set I and every family $P : I \to \mathrm{Prop}$:*

1. The cofinite filter is the meet of the free ultrafilters *(`mem_cofinite_iff_forall_free`, `iSup_free_eq_cofinite`): a set is cofinite exactly when every free ultrafilter contains it.*
2. What every free One affirms is what holds cofinitely *(`forall_free_ultraTrue_iff`):* $\big(\forall \mathcal{U}$ *free,* UltraTrue$(\mathcal{U}, P)\big)$ *iff* $\{i : \neg P\, i\}$ *is finite.*
3. Some free One affirms what holds infinitely often *(`exists_free_ultraTrue_iff`).*
4. The free dial *(`freeOpen_iff`): two free Ones disagree about P exactly when P holds at infinitely many coordinates and fails at infinitely many.*
5. Łoś forms *(`forall_free_uprod_iff`, `exists_free_uprod_iff`, `freeOpen_uprod_iff`): for a family of first-order structures A_i and a sentence φ, φ holds in every free ultraproduct iff it holds at all but finitely many A_i; in some free ultraproduct iff at infinitely many; and the free ultraproducts disagree iff both sets are infinite.*

Controls on $\mathbb{N}$*: "nonzero" holds in every free One (`forall_free_ne_zero`), "zero" in none (`not_exists_free_eq_zero`), and "even" is open among them (`freeOpen_even`).*

Part 5 composes the wrapper form of Łoś's theorem of the first essay (`ultraTrue_iff_uprod`, `Metaphysics/UltrainfinitismLosTransfer.lean`) with parts 2–4.

Reading 15 (effable and chosen)*.* The cofinite filter is defined without choice; the free ultrafilters are not. Bridge 14 says that the meet of all the choices is exactly the choice-free part. So the indexed One divides cleanly: what holds *eventually* can be effed, and is affirmed by every free perspective; everything else depends on which free ultrafilter is taken, which is a choice. The first essay's credal width measured disagreement among perspectives; here the free perspectives disagree exactly on that remainder (part 4).

3.2 B2: no symmetric One

Bridge 16 (no symmetric choice of a free One; `Metaphysics/UltrainfiniteSymmetry.lean`, `Metaphysics/UltrainfiniteSymmetryObserver.lean`).

1. The shared form *(`select_ne_of_fixed`, no axioms): if a selection—a function, hence deterministic—commutes with a symmetry, then at a point the symmetry fixes, the selection is neither member of a pair the symmetry exchanges.*

2. Ultrafilters *(`map_ne_of_preimage_eq_compl`): a map that carries a set onto its complement moves every ultrafilter. In particular the swap of* $2k$ *and* $2k+1$ *moves every ultrafilter on* $\mathbb{N}$ *(`map_evenOddSwap_ne`), so* no ultrafilter on $\mathbb{N}$ is fixed by every permutation *(`no_permutation_invariant_ultrafilter`).*

3. Control *(`map_swap_eq_of_isFree`, `hyperfilter_swap_control`): every free ultrafilter is fixed by every transposition, hence by every permutation that moves finitely many points.*

4. The observer instance *(`no_equivariant_selection_of_shared`): an observer's deterministic selection that is equivariant under the stated renaming action, with no choice fixed in advance, selects neither member of a pair exchanged by an automorphism of its record. That this is literally the observer library's theorem `no_equivariant_selection` (`Step/ObserverChoices.lean`) is checked by `rfl` (`no_equivariant_selection_of_shared_eq`).*

The two obstructions in part 2 and part 4 are instances of one form, and neither is an instance of the other: the ultrafilter theorem concerns a selection every ultrafilter makes ("the half of $\{S, S^c\}$ it contains"), so it concludes that no ultrafilter is fixed at all, while the observer theorem concerns records and their renamings. What a free ultrafilter decides is invisible to finite changes (part 3); only an infinite symmetry moves it. Beyond the formal statement, ZF without choice does not prove that a free ultrafilter on $\mathbb{N}$ exists [17]: the free One is not merely unsymmetric, it is unconstructed.

3.3 B3: $\mathcal{P}(\mathbb{N})$/fin

Bridge 17 ($\mathcal{P}(\mathbb{N})$/fin; `instBooleanAlgebraGerm`, `coe_eq_bot_iff`, `isGunky_setModFinite`, `perfectSpace_stoneSpace_setModFinite`, `Metaphysics/UltrainfiniteSetModFinite.lean`). *Germs along a filter of functions into a Boolean algebra form a Boolean algebra, computed pointwise. For the cofinite filter on* $\mathbb{N}$ *and `Prop` this is* $\mathcal{P}(\mathbb{N})$/fin*, the subsets of* $\mathbb{N}$ *with finite differences ignored. Its zero is the class of all finite sets; it has no atoms, since every infinite set splits into its members of even and of odd rank; and its Stone space is perfect.*

Remark 18 (why a bridge). $\mathcal{P}(\mathbb{N})/\mathrm{fin}$ is the clearest image of the thesis inside classical mathematics: infinitely many empties (every finite set) identified into one zero, no atom, no isolated point. Classically its Stone space is $\mathbb{N}^* = \beta\mathbb{N} \setminus \mathbb{N}$, the space of free ultrafilters on $\mathbb{N}$ [14, 40]. But it is built from the subsets of $\mathbb{N}$, hence from the finite, and its zero is defined as the finite sets. It is therefore a bridge in the sense of Principle 1. The identification of its Stone points with the free ultrafilters is classical and is not proved in the development (Section 9).

4 Three senses of "without normal forms"

Ultrainfinitist slogans often say that wholes "have no normal form". An objection stands against the naive version, and it is worth stating first.

The objection. In a final coalgebra, bisimilar behaviours are *equal* [34, 3]. So each stream in Cantor space, each hyperset, each rational tree is already canonical as a denotation. The development agrees: rational trees are equal exactly when their presenting graphs are bisimilar (`unfold_eq_iff`, `Step/RationalTrees.lean`); unification over them is unitary, with a canonical solved form built from the classes of a closure (`classTree`, `Step/RationalUnification.lean`) and computed as a finite graph (`solvedGraph`, `Step/RationalSolvedForm.lean`). "No normal form" cannot simply mean "no canonical denotation".

Three precise senses survive. They concern *representatives*, not denotations.

4.1 No finite, well-founded representative

The leafless wholes of Theorem 4 have no finite term reading (`not_leafless_ofTerm`); the rational tree $f(f(f(\cdots)))$ solving $X = f(X)$ has none either. The candidate step of Section 7 makes the contrast operational: the firing that tells $X = f(X)$ is refused over finite trees at every configuration (`tellLoopFinite_refused`, `Step/RationalStep.lean`), admitted over rational trees from no knowledge (`tellLoop_admitted`), and its run ends in knowledge solved by $X := f(f(f(\cdots)))$ (`tellLoop_run`). The ω-limits of Theorem 25 are of the same kind: each iterate is finite, the limit is not.

4.2 No canonical representative without breaking a symmetry

Configurations that differ only by a private renaming of names form orbits. Picking one member per orbit is a selection, and Bridge 16 limits selections: no deterministic selection that commutes with renaming can pick a member of a pair that a symmetry exchanges. In the nominal setting, where names carry no order, an invariant choice of orbit representative must be fixed by every renaming, and in general there is none [31]. The development records the alternative directly: every equivalence has a canonical form *by choice* (`Canonicalizer.choose`,

`Step/Fold.lean`), and computing one is left to a profile. Canonical forms on a computer exist because its names are ordered; ordering the names is a choice that fixes the gauge. Even a minimal presentation of a rational tree is unique only up to renaming its nodes, so making it a single datum asks again for an order on names.

4.3 No normal form at self-containing types

The third sense is the sharpest, and it is a theorem about logic rather than data. Lawvere's fixed-point theorem [27, 43] holds for types in the following form.

Theorem 19 (Lawvere for types; `lawvere_fixedPoint`, `diagonalFix_spec`, `Metaphysics/UltrainfiniteLawvere.lean`)**.** *If $e : A \to (A \to B)$ is surjective, every $f : B \to B$ has a fixed point; with a section of e the fixed point is the term $e(sg)(sg)$ with $g = \lambda a.\, f(e\,a\,a)$, the shape of $(\lambda x.\, f(xx))(\lambda x.\, f(xx))$.* `lawvere_fixedPoint` *depends on no axioms. Corollaries:*

1. Cantor *(*`not_surjective_prop`*,* `cantor_surjective`*): negation has no fixed point in* `Prop`*, so no $A \to (A \to \mathrm{Prop})$ is surjective;*
2. Curry *(*`nonempty_of_surjective`*): a codomain reached by a point-surjection is inhabited, and (*`subsingleton_of_surjective`*) has at most one element;*
3. the self-containing type is a One *(*`unique_of_equiv_arrow`*): a type $X \simeq (X \to X)$ has exactly one element;* `PUnit` *is such a type and* `Bool` *is not (*`punitEquivArrow`*,* `not_nonempty_bool_equiv_arrow`*);*
4. Tarski *(*`no_truth_predicate`*, no choice): there is no* $\mathrm{Tr} : X \to \mathrm{Prop}$ *with a substitution through which every predicate on X is defined by a code.*

Theorem 20 (rational types collapse; `omega_typable`, `omega_no_normal_form`, `cyclic_type_breaks_normalization`, `Step/RationalTypeControl.lean`; `rarrow_curryType`, `ArrowModel.unique_carrier`, `occurs_check_keeps_models`, `Metaphysics/UltrainfiniteLawvereTypes.lean`)**.**

1. *Over finite types, self-application $\lambda x.\, xx$ is untypable (*`selfApplication_untypable`*). Over rational types, $\Omega = (\lambda x.\, xx)(\lambda x.\, xx)$ is typable at the rational type $T = T \to T$, and Ω has no normal form: its only reduct is itself.*
2. *Every rational type b has a Curry type C with $C = C \to b$.*
3. Every arrow model is trivial*: in every interpretation of the rational types as types that reads the arrow as the function type up to equivalence, every rational type—$T = T \to T$ included—has exactly one element.*
4. Control*: the finite types have an arrow model with a two-element base type (*`FiniteArrowModel.bool_nontrivial`*).*

Here $T = T \to T$ is exactly Lawvere's hypothesis: T maps onto T^T. So the non-normalizing term and the collapse of models are two faces of one diagonal. This is why the rational-tree fibre of the candidate step admits cyclic *data* but keeps the occurs check for *types*: a self-containing type logic inhabits every type and collapses every model in types. *Data may lack normal forms; the logic that reads it cannot contain itself.*

The collapse in part 3 decides equality classically and is a fact about types read as sets of all functions. In domain-theoretic models a nontrivial $D \cong [D \to D]$ exists, because the arrow there denotes only the continuous functions [35]. The theorem is therefore about the type discipline that uses the full function space, which is the one a proof assistant's logic uses.

4.4 The objection answered

The objection is right about denotations and is in fact the thesis. A final coalgebra makes the *whole* canonical: identity is behaviour, and behaviour is infinite. What is not canonical is any finite *presentation*: one rational tree has infinitely many presenting graphs, related by bisimulation; one orbit has no invariant representative; and at a self-containing type the logic that would read a normal form collapses. The three senses are three ways in which the finite fails to be the measure of the whole, while the whole is perfectly determinate.

5 The hierarchy of effability

What can a finite presentation do? Table 2 orders the grades. Each rung names the certificate that effs an object at that rung, and, where the development has one, the anchor that makes the certificate precise.

Rung	Certificate	Anchor or source
finished observation	a search that finished	status `finished`; `run_eq_of_finished`, `refuted_sound` (`Step/SearchSpace.lean`)
rational	a finite graph	`IsRational`, `rational_iff_unfold`; equality decided to a bounded depth, `unfold_eq_iff_agree`
computable	a program	Turing [39]
left-c.e.	answers that only grow	`run_mono`; Chaitin's Ω [10]
limit (Δ^0_2)	conjectures that settle	the first essay's §(A) [1]; Gold, Putnam, Shoenfield [21, 33, 36]
definable	a name, not an approximation	e.g. the second jump $\varnothing''$
arbitrary	none	ineffable relative to a fixed language

Table 2: The hierarchy of effability: finished < rational < computable < left-c.e. < limit < definable < arbitrary. Each inclusion is strict; the witnesses are standard (an infinite rational tree; a computable non-periodic stream; Ω; $1 - \Omega$; $\varnothing''$; a counting argument).

5.1 The rungs and the outcomes of search

The lower rungs are tied to an algebra of bounded-search outcomes. An outcome is a multiset of answers with a status, *finished* (the search looked everywhere) or *cut off* (it ran out of budget). It reads as one of three verdicts: *commit* (some answer), *refuted* (finished with no answers), *suspended* (cut off with no answers). A search space unfolds each state into alternatives, and the budgeted run at depth n reports the answers at depth at most n.

Theorem 21 (outcomes of bounded search; `Step/SearchOutcome.lean`, `Step/SearchSpace.lean`, `Step/SearchControls.lean`)**.**

1. Incomplete is never none *(`verdict_ne_refuted_of_cutOff`): a cut-off outcome never reads refuted. Definite verdicts are stable as the budget grows (`verdict_eq_of_le`).*

2. More budget only adds *(`run_mono`), and a finished run is the run at every larger budget (`run_eq_of_finished`).*

3. Refutation needs finished*: a run is cut off exactly when some state lies at depth at least n (`status_run_eq_cutOff_iff`), and a refuted readout is sound for the unbounded semantics (`refuted_sound`).*

4. Collect stays unknown *(`collect_unique`): on an inhabited answer type, the only monotone map that collects finished outcomes to their answer multisets sends every cut-off outcome to* unknown*. A choice with any cut-off branch collects to unknown (`collect_add_cutOff`).*

5. Control *(`never_finishes`): on a left-recursive space no policy ever finishes, at any budget.*

Read against the hierarchy: what converges under a growing budget is the set of answers found, a Σ_1^0 quantity, which is the left-c.e. shape. Refutation and totality are Π_1^0 and need a *finished* certificate; without one, part 1 forbids reading absence as refutation, and part 4 forbids collecting an infinite search into a value. Within this algebra a verdict changes at most once, from suspended to definite. The limit rung allows finitely many changes of mind [21, 33]. That is the situation of the first essay's conjecturer, whose verdict on a conjecture may change as more sizes are sampled: a conjecture whose verdict changes only finitely often settles in the limit, while one whose truth is ultrafilter-ambiguous in the tail need not settle at all.

5.2 Chalmers's "that's all" clause as the finished certificate

Chalmers's scrutability base includes, beside physical, phenomenal and indexical truths, a "that's all" clause: the world contains no more than what is needed to make the rest true [12, 11]. Negative truths—there are no ghosts—are scrutable only with it.

Reading 22 (that's all = finished). The "that's all" clause plays exactly the role of the *finished* status. "No answers" is a truth only with `finished` (Theorem 21, parts 1 and 3), as "no ghosts" is a truth only with the clause. The first essay already proved the concept-formation version of this correspondence: a concept is "that's all" exactly when its gate-verdict family is precise (`thatsAllConcept_iff_preciseFamily`, `Metaphysics/DialWeld.lean`). For a search whose tree is infinite the clause can never be issued by inspection: collecting all answers is defined exactly when some budget finishes (`all_dom_iff`, `Step/SearchFinish.lean`), and on the left-recursive control no budget does (Theorem 21, part 5).

5.3 Ideal scrutability versus effective effing

Scrutability is a priori entailment for an idealized reasoner [11]. Effing, as used here, keeps the budget: a finite, checkable presentation. The two come apart at Chaitin's halting probability Ω [10]. Each bit of Ω is fixed by arithmetical facts about which programs halt, so an ideal reasoner who knows those facts knows the bits. But Ω is only left-c.e.: it is approximable from below, its bits are not computable, and a consistent, effectively axiomatized theory determines only finitely many of them. Ω is ideally scrutable and not effable. This is a reading of a classical theorem, not a new result; it marks where the first essay's scrutability vocabulary and this essay's effability vocabulary diverge.

5.4 Caveats on the top rungs

Two familiar slogans need correcting.

"Normal numbers are not computable." False as stated. Champernowne's $0.123456789101112\ldots$ is normal in base ten and computable [13]; Turing gave an algorithm for absolutely normal numbers [7], and Becher and Figueira gave a computable absolutely normal number [6]. The true statements are that almost all reals, and almost all normal numbers, are not computable (the computable ones are countable, so of measure zero), and that normality itself is definable: read directly from Borel's definition [8], it is an arithmetical condition of the form $\forall\forall\exists\forall$.

"Most reals are undefinable." True only relative to a fixed countable language, seen from outside the model. Hamkins, Linetsky and Reitz built models of ZFC in which every real, indeed every set, is definable without parameters [22]. The top rung, "arbitrary", is therefore ineffable relative to a language and a vantage, not absolutely.

6 Two ineffabilities

The places where effing fails are of two kinds. One comes from *symmetry*: there is too little structure to single anything out. The other comes from *self-reference*: a description that reaches far enough to describe itself undermines itself. Figure 2 sets them side by side.

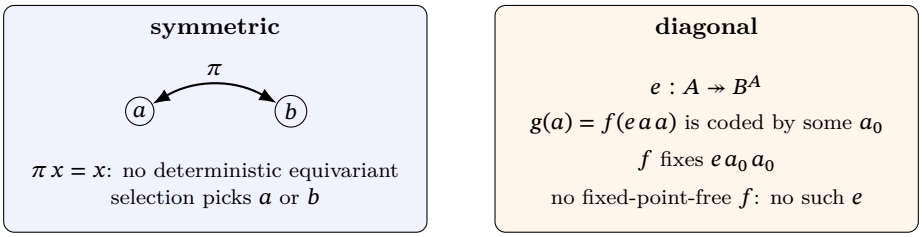


Figure 2: Two ineffabilities. Left: at a symmetric state no deterministic equivariant selection chooses between indiscernibles (Bridge 16). Right: a surjection onto its own exponent forces fixed points (Theorem 19).

6.1 Symmetric ineffability

A structural description fixes its realizers only up to automorphism. Lewis called the resulting ignorance *Ramseyan humility*: the Ramsey sentence of a complete physical theory is realized by any permutation of the fundamental properties that respects its structure, so we cannot know which property plays which role [28]. Bridge 16 is the formal core. At a state fixed by a symmetry, no deterministic selection that commutes with the symmetry can pick a member of an exchanged pair (`no_equivariant_selection`, `Step/ObserverChoices.lean`); and no ultrafilter on $\mathbb{N}$ survives the evens/odds swap (`no_permutation_invariant_ultrafilter`), so no free One is chosen symmetrically.

The scope of the theorem is exact. It excludes a *deterministic* selection that is equivariant for the stated action, with no choice fixed in advance. It does not exclude other ways of choosing: an asymmetry or extra state in the record can select, and a uniformly random choice can be equivariant as a distribution while each of its outcomes breaks the symmetry. The development models one such alternative. A selection may be a nondeterministic firing whose effect contract includes recording its choice: taking 5 and taking 6 from the sighted pair $\{5, 6\}$, each recorded, are trace equivalent, with records related by the exchange (`choice_recorded_identified`, `Step/ObserverChoices.lean`). That a choice is recorded is part of this firing's contract, not a consequence of `no_equivariant_selection`. Where a disclosure is recorded, it shrinks the symmetry group: the automorphisms of an extended record are those of the record that also fix the extension (`record_automorphisms_append`, `Step/Fold.lean`).

Local and global. A related gap concerns the order of quantifiers. Over a set of solutions of some knowledge, *pointwise* equivalence asks for a matching renaming at each solution; *uniform* equivalence asks for one renaming that matches at all of them (`pointwiseEquivalent_iff_stalks`, `moduloEquivalent_iff_global`, `Step/ObserverSections.lean`). A principal answer is an initial solution, and there the two coincide (`global_of_principal`); without one they come apart: two incomparable solutions each have a matching renaming, and no renaming matches at both (`local_sections_do_not_glue`). This is a local-versus-global gap in the order of quantifiers, "for each solution there is a renaming" against "there is a renaming for every solution". It is not contextuality in the sense of overlapping measurement contexts whose restrictions agree and fail to glue [2]: the two renamings in the example are not compatible even on the empty overlap. The same split separates "true in every free One" from "true in this One" in Bridge 14.

Reading 23 (choice among indiscernibles as free choice). One may read Bridge 16 as saying that no deterministic law can make a choice among indiscernibles at a symmetric state. A random draw can be symmetric in distribution, while each of its outcomes breaks the symmetry; recording the outcome is a further contract, not part of the theorem. Conway and Kochen's free will theorem has a similar shape—a response that is not a function of the information available to the responder [15]—but it is a parallel, not a consequence: nothing here derives it, and the present theorems concern selections and records, not physics.

Remark 24 (an interpretive aside). In the channelled *Law of One* material, free will is named the *first distortion* of the Law of One [16]. Read against the theorem above, the image is apt: no deterministic law selects at the undivided state, and each outcome of a selection breaks its symmetry. We note the parallel as interpretation only and draw nothing from it.

6.2 Diagonal ineffability

The second kind is Lawvere's. Theorem 19 is one statement whose instances are the classical limits of description: Cantor's theorem that no set maps onto its powerset [9] (`cantor_surjective`), Tarski's undefinability of truth [38] (`no_truth_predicate`), Curry's paradox read on types (`nonempty_of_surjective`), the impossibility of a transparent self-reflection (`transparent_reflection_impossible`, Theorem 10), and the collapse of self-containing types (Theorem 20). Gödel's incompleteness theorem [20] and Chaitin's bound on the bits of Ω a theory can determine [10] share the diagonal form [43]; they are cited here, not formalized.

The two kinds differ in what would remove them. Symmetric ineffability is removed by something beyond a deterministic equivariant law: a choice fixed in advance, an asymmetry or extra state, or a random draw whose outcome breaks the symmetry. Diagonal ineffability is removed by no choice at all: it is a bound on any total internal description, however chosen. The first is a shortage of structure, the second an excess of reach.

7 Transforms over transfinite entities

The thesis is about wholes; computation is about steps. This section shows that steps over infinite wholes behave as the thesis predicts: their limits are characterized by the whole, their conserved quantities survive passage to the hyperfinite, their silent loops close up to renaming, and a single cycle can grow without bound.

The candidate step, as far as needed. A state of the candidate step is a *space*—a finite set of occurrences, each carrying a labelled term—together with *knowledge*, an equation store over terms. A rule's *patterns* say what a firing may do, in five positions: it *takes* occurrences (consuming them), *reads* occurrences (without consuming them), *makes* new occurrences, *tells* equations to the knowledge, and is admitted only if its *guards* hold at every solution of the knowledge. A run is a word of admitted firings. The search combinators (`first`, `take`, `all`, `within`, `Step/SearchPolicy.lean`) explore runs under a budget, with the outcomes of Theorem 21. Over rational trees the first-order layer of the same step runs unchanged; only the meanings of "solvable", "asked" and "absent" change (Section 4.1).

7.1 D1: ω-limits of transforms of infinite trees

Let AgreeBelow(d, s, t) say that two trees have the same label at every path shorter than d: a finite observation. A run of trees *converges* to t when at every depth it is eventually equal to t below that depth. A transform Φ of whole trees is *productive* when inputs agreeing below d give outputs agreeing below $d + 1$.

Theorem 25 (ω-limits; `ConvergesTo.unique`, `Productive.existsUnique_fixed`, `productive_guard`, `tellLoop_unfolding_limit`, `Step/TransfiniteLimits.lean`)**.**

1. *Limits are unique.*
2. *A productive transform has exactly one fixed point, and from every start its run of iterates converges to it.*
3. *A guard makes a transform productive:* $t \mapsto h(t, \dots, t)$.
4. The rational step on $X = f(X)$*: the single firing that tells* $X = f(X)$ *ends in knowledge solved by* $X := f(f(f(\cdots)))$*; the unfolding run* X, $f(X)$, $f(f(X)), \dots$ *converges path-wise to that tree; and it is the only tree solving the equation.*
5. Controls*: the transform that flips the root between two constants is not productive and none of its runs converges (`not_convergesTo_rootFlip`, `not_productive_rootFlip`); the unguarded equation* $X = X$ *has more than one solution (`unguarded_not_unique`).*

Part 2 is the contraction principle for the depth ultrametric on trees [4], read path by path. Each iterate of the unfolding run is a finite tree, but the limit is not assembled from them: it is *characterized* as the unique fixed point of a transform of whole trees, and the iterates are its observations below growing depths. That is Principle 1 in operational form.

7.2 D2: hyperfinite runs

A *conserved charge* of a step is a value on states together with what each firing takes and makes, balanced at every admitted firing; summed along a run, the books balance (`run_balance`, `Step/Charge.lean`), and a run that does not take what it makes cannot fold back to its start (`not_fold_of_unbalanced`). Occurrence weights give such a charge for every step (`weightCharge`). A *hyperfinite run* is a family of runs, one at each index, read modulo an ultrafilter $\mathcal{U}$: its length, its potentials at start and end, and what it takes and makes are germs along $\mathcal{U}$.

Bridge 26 (charges through an ultrapower; `HyperRun.balance`, `HyperRun.balance_weight`, `germ_ne_iff`, `HyperRun.eventually_not_fold`, `HyperRun.taken_eq_made_of_eventually_fold`, `HyperRun.const_lt_length`, `cofinite_germ_ne_not_eventually_ne`, `Step/TransfiniteUltrapower.lean`)**.**

1. Conservation*: for every conserved charge, the end potential plus everything taken equals the start potential plus everything made, in the ultrapower; in particular for the occurrence weights of every step.*

2. The fold filter transfers*: for a charge invariant under the transport, with values in a left-cancellative monoid, if the hyperfinite run does not take what it makes, then $\mathcal{U}$-almost every coordinate run fails to fold back to its start; conversely, if $\mathcal{U}$-almost every coordinate folds, the hyperfinite run takes what it makes. The proof uses that $\mathcal{U}$ is an ultrafilter: an inequality of germs is an inequality at $\mathcal{U}$-almost every index (Łoś for a negated equation).*

3. Infinite length*: if $\mathcal{U}$ is free and, for every n, only finitely many coordinate runs have length at most n, then the length exceeds every standard number.*

4. Control*: over the cofinite filter, which is not an ultrafilter, germs can differ without differing eventually (the indicator of the evens against 0). A reduced power does not suffice; the ultrafilter is needed.*

This is labelled a bridge because a hyperfinite run is assembled from finite coordinate runs, in the manner of Nelson's radically elementary probability [30]. A germ has no coordinates of its own; what can be read from it is what holds on a set in $\mathcal{U}$.

7.3 Autopoiesis as the quiet fold

A search folds a descendant onto an ancestor when the two lie in one state class. When the observer's record is carried in the state (the escape-relative transport), folding is restrictive.

Theorem 27 (the quiet fold; `escape_fold_quiet`, `escape_fold_iterates_quiet`, `Step/Fold.lean`)**.** *In the escape-relative transport, a run that ends in the class of its start recorded nothing, and every isomorphism closing it renames by an automorphism of the start's record. Every replay of such a loop records nothing.*

Reading 28 (autopoiesis). Maturana and Varela defined an autopoietic system as one whose organization is invariant while its components are continually replaced [29]. Theorem 27 has that shape: the organization returns (the state class of the start), while the parts may be renamed (the closing isomorphism may move occurrence identities), and the renaming is invisible to the environment (an automorphism of what was disclosed). A loop that discloses cannot close this way: disclosure shrinks the folding group (`record_automorphisms_append`). We read self-maintenance as the quiet fold; the theorem itself concerns runs, records and renamings.

7.4 Unbounded growth from one cycle

Theorem 29 (the affine pump; `affine_pump`, `Step/FoldPump.lean`)**.** *Suppose a word of firings w runs from a configuration C and ends in $C \otimes \delta$ up to equivalence, where every firing of w is query-free (equation guards only, no predicate query), δ carries no knowledge, δ lies beside C, and for every finite*

set of identities there is a label-keeping renaming that maps the occurrences of C onto themselves and moves what w makes off that set. Then for every n there is an executable word ending in $C \otimes E$, where E carries no knowledge, lies beside C, and has n times the labels of δ. Controls *(`producer_pumps`, `absent_guard_stops`): a producer from the empty configuration pumps to n occurrences for every n; a rule that makes b while requiring b absent fires once and is then refused, under every renaming.*

A positive balance alone does not give unbounded growth: the refused control makes one occurrence and takes none, yet fires only once. Every accumulated frame must keep the cycle admissible, and the hypotheses say exactly that. Absence and count guards break it, because they are not blind to what the cycle has already made.

7.5 The irony, kept explicit

Every presentation in this section is finite: a rule with finitely many patterns, a finite graph, a finite word of firings, a Lean file of a few hundred lines. What these presentations present need not be finite: an infinite tree characterized as a fixed point, a run whose hyperfinite length exceeds every number, an unbounded family of configurations from one cycle, an uncountable group of automorphisms of one whole. This is the first essay's use-versus-mention reply, now with the mechanisms named. A finite presentation is an observation scheme; it denotes a whole by saying how the whole may be observed.

8 Effing the ineffable's boundary

The second half of this essay has a shape that is easy to miss. Its two ineffabilities are not gestures toward the unsayable; they are *theorems*, finite and checked, and some of them use no axioms at all. `lawvere_fixedPoint` depends on nothing; `select_ne_of_fixed` depends on nothing; `no_truth_predicate` and the sinkless collapse use no choice. Each says precisely where a class of finite presentations must stop: no deterministic equivariant selector at a symmetric state, no surjection onto an exponent with a fixed-point-free endomap, no collection of an unfinished search, no refutation without a finished certificate, no nontrivial model of a self-containing type. The boundary of effing is itself effed.

This is not a paradox, and it is not the ladder that is thrown away [42]. Priest showed how the diagonal limits of thought share one schema [32]; Lawvere had already shown that the classical diagonal arguments are instances of a single fixed-point theorem [27, 43]. Here the same holds for the symmetric limit: one short form, `select_ne_of_fixed`, carries both the ultrafilter obstruction and the observer obstruction. The ultrainfinitist position thereby gains a stable shape. The infinite whole is primary (Principle 1); finite observation reaches it by prefixes, budgets, coordinates and clopens; and the two places where obser-

vation cannot reach are marked by finite theorems. We do not say the ineffable. We say, exactly, where saying ends.

What remains is foundational work, listed in the next section: the leafless wholes as a final coalgebra, the Stone points of $\mathcal{P}(\mathbb{N})/\mathrm{fin}$, and a genuine ultrapower of the step.

9 Honest scope

Status	Items
Proved (named anchors)	sinkless collapse (Thm. 2); leafless plurality (Thm. 4); inner diversity (Thm. 6); no locale points (Thm. 8); finite shadows, continuity as finite-prefix dependence, and finite-resolution observers (Thm. 12, Section 2.6); opaque atoms (Thm. 10); Lawvere for types and rational types (Thms. 19–20); bounded-search outcomes (Thm. 21); ω-limits (Thm. 25); the quiet fold (Thm. 27); the affine pump (Thm. 29).
Bridges (proved; built from finite pieces)	B1 the cofinite meet and its Łoś forms (Bridge 14); B2 the symmetry obstructions in a form shared with the observer theorem (Bridge 16); B3 $\mathcal{P}(\mathbb{N})/\mathrm{fin}$ (Bridge 17); D2 hyperfinite runs (Bridge 26). Free ultrafilters come from choice.
Readings (interpretation)	the Law of Oneish by structure; plurality by carving; connected, not contractible; the pointless One; the empty as an outside view; effable and chosen; that's all = finished; ideal scrutability versus effing; free choice among indiscernibles and the Conway–Kochen parallel; the aside on the *Law of One*; autopoiesis as the quiet fold; the computable as a shadow.
Literature, not formalized	the strictness of the effability hierarchy above the rational rung; Chaitin's Ω; computable normal numbers; pointwise definable models; $\mathrm{St}(\mathcal{P}(\mathbb{N})/\mathrm{fin}) \cong \mathbb{N}^*$; ZF without choice and free ultrafilters; Gödel's theorem; a general Stone space is the inverse limit of its finite clopen partitions; computable maps on infinite objects are continuous, in the uniform oracle form.
Open	A3: leafless trees as a final coalgebra, so the core objects are characterized by a universal property; the inverse-limit presentation of a general Stone space by its finite shadows, and the uniform oracle form of "computable implies continuous" (Section 2.6); the Stone points of $\mathcal{P}(\mathbb{N})/\mathrm{fin}$ are exactly the free ultrafilters on $\mathbb{N}$; admission in an ultrapower of the step iff admission at $\mathcal{U}$-almost every coordinate, which needs a genuine ultrapower of the first-order step; full perfectness, that every leafless tree has a distinct neighbour at every depth (Theorem 4, part 4, gives only some pair at each depth); ω-limits for graph systems beyond $X = f(X)$; the symmetry obstruction for every infinite type; the coalgebraic, non-well-founded form of opaque atoms.

Table 3: What is proved, what is a bridge, what is a reading, what is cited, and what is open.

Where finitary scaffolding remains. Principle 1 holds at the object level of several results while the metatheory and the presentations stay finitary.

Result	"No finite parts" at the object level	Finitary scaffolding
sinkless collapse	no member of Ω's closure is empty; Ω is characterized by a property	hypersets are quotients of graphs, which may be finite; the closure is reached by finite chains
leafless, inner diversity	no leaves; identity carried as structure	paths are finite lists; rational trees are counted through finite codes; equality of permutations is Lean's equality, not a homotopical identity
Lawvere, rational types	a statement about types	the collapse to one element decides equality classically; it is a fact about full function spaces (Scott's $D \cong [D \to D]$ is nontrivial)
ω-limits	the limit is characterized as the unique fixed point	its construction reads finite iterates
B1, B3, D2	—	bridges, defined by finiteness over coordinates; free ultrafilters by choice
opaque atoms	"empty" is a property of the view	a finitary, well-founded syntactic tier

Table 4: Where "no finite parts" holds, and where the scaffolding is still finite.

No claim in this essay is stated beyond what its cited anchor proves. The readings are offered as readings; the bridges are used as bridges.

A Formalized statements (Lean 4)

The statements below are transcribed from the source, with Unicode notation spelled in ASCII: `forall`, `exists`, `<->`, `!=`, `in`, `<=`, `~=` for an equivalence of types, `forall^f` for "eventually", `|=` for satisfaction, `alpha + sigma` for a sum type, `<A, s>` for an anonymous constructor, `*s` for the n-fold sum, and Greek letters by name. Implicit arguments are sometimes omitted. None of the modules contains a hole marker.

Sinkless collapse (`Metaphysics/UltrainfiniteSinkless.lean`).

```
/-- y is in the transitive closure of x, x itself included. -/
def InClosure (x y : HSet) : Prop := Relation.ReflTransGen (fun a b
    => b in a) x y

/-- Hereditarily nonempty: every member of the closure, itself
    included, has a member. -/
```

```
def HereditarilyNonempty (x : HSet) : Prop := forall y, InClosure x y
     -> exists z, z in y

/-- Sinkless collapse: two hereditarily nonempty hypersets are equal.
     -/
theorem eq_of_hereditarilyNonempty (hx : HereditarilyNonempty x)
    (hy : HereditarilyNonempty y) : x = y

/-- The only hereditarily nonempty hyperset is Omega. -/
theorem hereditarilyNonempty_iff_eq_quineAtom :
    HereditarilyNonempty x <-> x = quineAtom

/-- A graph pictures Omega at a node exactly when no sink is
    reachable from it. -/
theorem decorate_eq_quineAtom_iff_sinkless {a : alpha} :
    decorate r a = quineAtom <-> forall b, Relation.ReflTransGen r a
        b -> exists c, r b c

/-- With a sink, plurality appears. -/
theorem sink_plurality :
    (emptyset : HSet) != {emptyset} /\ not (HereditarilyNonempty
        emptyset) /\
      not (HereditarilyNonempty {emptyset})
```

Leafless trees and inner diversity (`Metaphysics/UltrainfiniteLeafless.lean`, `Metaphysics/UltrainfiniteInnerDiversity.lean`).

```
/-- A leafless tree: well formed, and every label present has a child
    . -/
def Leafless (t : Raw sigma) : Prop := WF t /\ forall p h, t p = some
     h -> 0 < h.arity

theorem Leafless.infinite_support (l : Leafless t) : {p | t p != none
    }.Infinite
theorem not_leafless_ofTerm (u : Term sigma) : not (Leafless (ofTerm
    u))

theorem not_countable_setOf_leafless (apart : f != g) (pf : 0 < sigma
    .functionArity f)
    (pg : 0 < sigma.functionArity g) : not {t : Raw sigma | Leafless
        t}.Countable

theorem countable_leafless_rational [Countable sigma.functionSymbols]
     :
    {t : Raw sigma | Leafless t /\ IsRational t}.Countable

theorem exists_ne_agree_below (apart : f != g) (pf : 0 < sigma.
    functionArity f)
    (pg : 0 < sigma.functionArity g) (d : Nat) :
    exists s t : Raw sigma, Leafless s /\ Leafless t /\ s != t /\
      forall p : List Nat, p.length < d -> s p = t p

theorem existsUnique_leafless_of_subsingleton [Subsingleton sigma.
    functionSymbols]
    (f : sigma.functionSymbols) (pf : 0 < sigma.functionArity f) :
```

```
    exists! t : Raw sigma, Leafless t

/-- One whole, many self-identifications. -/
theorem inner_diversity :
    not (Countable treeAut) /\ (forall p, decorate Child p =
        quineAtom) /\
      forall pi : HSet ~= HSet, (forall x y, y in x <-> pi y in pi x)
          ->
        pi quineAtom = quineAtom
```

Points and opaque atoms (Mettapedia/SetTheory/CarveOuts/HeytingValued/Gunky.lean, Mettapedia/SetTheory/ColoredOpaqueUniverses.lean).

```
/-- Every point of a complete Boolean algebra is generated by an atom
    . -/
theorem exists_atom_of_point (p : Point B) :
    exists a, IsAtom a /\ forall h, p.holds h <-> a <= h

/-- An atomless complete Boolean algebra has no point. -/
theorem isEmpty_point_of_isGunky (hg : IsGunky B) : IsEmpty (Point B)

/-- The double-negation part has no two-valued reading. -/
theorem isEmpty_point_regular_opens [T2Space X]
    (noIsolated : forall x : X, not (IsOpen ({x} : Set X))) :
    IsEmpty (Point (Regular (Opens X)))

/-- Opacity: an atom exposes no members, whatever its payload holds.
    -/
theorem atom_memberless (sees : V.Sees c d) (payload : WfObject V d)
    (x : WfObject V c) :
    not (MemberOf x (.atom sees payload))

/-- Reference retains identity: distinct payloads give distinct atoms
    . -/
theorem atom_injective (sees : V.Sees c d) :
    Function.Injective (WfObject.atom (V := V) (c := c) sees)
```

Bridges (Metaphysics/UltrainfiniteCofinite.lean, Metaphysics/UltrainfiniteCofiniteLos.lean, Metaphysics/UltrainfiniteSymmetry.lean, Metaphysics/UltrainfiniteSymmetryObserver.lean, Metaphysics/UltrainfiniteSetModFinite.lean).

```
/-- A free ultrafilter: the principal ultrafilter of no point. -/
def IsFree (U : Ultrafilter I) : Prop := forall i, U != pure i

theorem mem_cofinite_iff_forall_free {s : Set I} :
    s in cofinite <-> forall U : Ultrafilter I, IsFree U -> s in U

theorem forall_free_ultraTrue_iff (P : I -> Prop) :
    (forall U : Ultrafilter I, IsFree U -> UltraTrue U P) <-> {i |
        not (P i)}.Finite
```

```
theorem freeOpen_iff (P : I -> Prop) :
    (exists U V : Ultrafilter I, IsFree U /\ IsFree V /\ UltraTrue U
        P /\
        not (UltraTrue V P)) <->
      {i | P i}.Infinite /\ {i | not (P i)}.Infinite

theorem forall_free_uprod_iff (phi : Sentence L) :
    (forall U : Ultrafilter I, IsFree U -> (Uprod A U) |= phi) <->
      {i | not (A i |= phi)}.Finite

/-- The shared form (no axioms). -/
theorem select_ne_of_fixed (act : X -> X) (act' : Y -> Y) (select : X
     -> Y)
    (equivariant : forall x, select (act x) = act' (select x)) (fixed
         : act x = x)
    (apart : a != b) (exchanges : act' a = b /\ act' b = a) :
    select x != a /\ select x != b

theorem no_permutation_invariant_ultrafilter :
    not (exists U : Ultrafilter Nat, forall pi : Equiv.Perm Nat, U.
        map pi = U)

theorem map_swap_eq_of_isFree [DecidableEq alpha] {U : Ultrafilter
    alpha}
    (free : IsFree U) (a b : alpha) : U.map (Equiv.swap a b) = U

theorem no_equivariant_selection_of_shared_eq :
    @no_equivariant_selection_of_shared = @AnnotatedRuns.
        no_equivariant_selection := rfl

abbrev SetModFinite := Germ (cofinite : Filter Nat) Prop
theorem coe_eq_bot_iff (S : Nat -> Prop) : (S : SetModFinite) = bot
    <-> {n | S n}.Finite
theorem isGunky_setModFinite : IsGunky SetModFinite
theorem perfectSpace_stoneSpace_setModFinite : PerfectSpace (
    StoneSpace SetModFinite)
```

Lawvere and rational types (Metaphysics/UltrainfiniteLawvere.lean, Metaphysics/UltrainfiniteLawvereTypes.lean, Step/RationalTypeControl.lean).

```
/-- Lawvere's fixed-point theorem for types (no axioms). -/
theorem lawvere_fixedPoint (e : A -> A -> B) (surjective : Function.
    Surjective e)
    (f : B -> B) : exists b, f b = b

/-- The self-containing type is a One. -/
theorem unique_of_equiv_arrow {X : Sort u} (e : X ~= (X -> X)) :
    Nonempty (Unique X)

/-- No internal truth predicate. -/
theorem no_truth_predicate (Tr : X -> Prop) (sub : X -> X -> X) :
    not (forall P : X -> Prop, exists c, forall x, Tr (sub c x) <-> P
         x)
```

```
theorem omega_typable : HasType rarrow [] omega (loop .arrow)
theorem omega_no_normal_form (t : Lam) (reached : Relation.
    ReflTransGen Lam.Beta omega t) :
    exists u, Lam.Beta t u

/-- An arrow model: a type for each rational type, the arrow read as
    the function type. -/
structure ArrowModel where
  carrier : RTree ctlSig -> Sort u
  arrow : forall a b, carrier (rarrow a b) ~= (carrier a -> carrier b
      )

theorem rarrow_curryType (b : RTree ctlSig) : rarrow (curryType b) b
    = curryType b
theorem ArrowModel.unique_carrier (b : RTree ctlSig) : Nonempty (
    Unique (M.carrier b))

theorem occurs_check_keeps_models :
    (exists M : FiniteArrowModel, exists a b : M.carrier (.const true
        ), a != b) /\
      forall (M : ArrowModel) (b : RTree ctlSig), Nonempty (Unique (M
          .carrier b))
```

Bounded search (Step/SearchOutcome.lean, Step/SearchSpace.lean).

```
inductive Status where | finished | cutOff
structure Outcome (alpha : Type u) where
  answers : Multiset alpha
  status : Status

theorem verdict_ne_refuted_of_cutOff {o : Outcome alpha} (cut : o.
    status = .cutOff) :
    o.verdict != .refuted

theorem collect_unique [Nonempty alpha] (c : Outcome alpha -> Outcome
     (Multiset alpha))
    (monotone : Monotone c)
    (onFinished : forall A : Multiset alpha, c <A, .finished> = pure
        A) : c = collect

theorem run_mono (x : alpha + sigma) : Monotone fun n => S.run n x

theorem status_run_eq_cutOff_iff (n : Nat) (x : alpha + sigma) :
    (S.run n x).status = .cutOff <->
      exists w t, S.item x w = some (.inr t) /\ n <= w.length

theorem refuted_sound (h : (S.approx n x).verdict = .refuted) :
    forall w a, S.item x w != some (.inl a)
```

Transforms (Step/TransfiniteLimits.lean, Step/TransfiniteUltrapower.lean, Step/Fold.lean, Step/FoldPump.lean).

```
def AgreeBelow (d : Nat) (s t : Raw sigma) : Prop := forall p : List
    Nat, p.length < d -> s p = t p
def ConvergesTo (run : Nat -> Raw sigma) (t : Raw sigma) : Prop :=
  forall d, exists N, forall n, N <= n -> AgreeBelow d (run n) t
def Productive (Phi : Raw sigma -> Raw sigma) : Prop :=
  forall d s t, AgreeBelow d s t -> AgreeBelow (d + 1) (Phi s) (Phi t
      )

theorem Productive.existsUnique_fixed (t0 : Raw sigma) :
    exists! a, Phi a = a /\ forall t : Raw sigma, ConvergesTo (fun n
        => Phi^[n] t) a

theorem HyperRun.balance :
    R.endPotential U Q + R.taken U Q = R.startPotential U Q + R.made
        U Q

theorem HyperRun.eventually_not_fold [IsLeftCancelAdd A] {T :
    Transport P}
    (invariant : Q.Invariant T) (unbalanced : R.taken U Q != R.made U
         Q) :
    forall^f i in U, not (T.Same (R.start i) (R.run i).last)

theorem HyperRun.const_lt_length (free : (U : Filter I) <= cofinite)
    (unbounded : forall n : Nat, {i | (R.run i).word.length <= n}.
        Finite) (n : Nat) :
    (n : Germ U Nat) < R.length U

theorem cofinite_germ_ne_not_eventually_ne :
    (evenIndicator : Germ (cofinite : Filter Nat) Nat) != (fun _ =>
        0) /\
      not (forall^f n in cofinite, evenIndicator n != 0)

theorem escape_fold_quiet {x : Escaped P} (r : D.step.Run x)
    (fold : (T.escaping natural D).Same x r.last) :
    r.last.record = x.record /\
      forall i, (T.toStateIso i.base).rename in Record.automorphisms
          x.record

theorem affine_pump (free : forall f in w, QueryFree f) (runs :
    Executable A label C w)
    (ends : SameUpTo A (follow label C w) (tensor C delta))
    (spatial : A.Equivalent delta.equations []) (beside : Disjoint C.
        present delta.present)
    (supply : forall F : Finset O, exists pi : O ~= O, (forall o,
        label (pi o) = label o) /\
      C.present.map pi.toEmbedding = C.present /\
      Disjoint (madeBy (w.map (Firing.rename pi))) F) :
    forall n : Nat, exists u E, Executable A label C u /\
      SameUpTo A (follow label C u) (tensor C E) /\ A.Equivalent E.
          equations [] /\
      Disjoint C.present E.present /\
      E.present.val.map label = n *s delta.present.val.map label
```

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